Project Title

“Data Analyst Job Market Intelligence Dashboard”

Analyzing Hiring Trends, Skill Demand, and Salary Insights in the Data Analyst Job Market

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Report Prepared By

Meghana M

Bachelor of Engineering
Computer Science and Engineering

Tools & Technologies Used

Snowflake
Julius AI
Power BI
Data Analysis Expressions
Data Visualization

ABSTRACT

The rapid growth of the data-driven economy has significantly increased the demand for data analytics professionals across industries. Understanding hiring patterns, required skills, salary trends, and geographic demand is essential for job seekers, training institutions, and organizations planning their workforce strategies.

This project presents a **Data Analyst Job Market Intelligence Dashboard** developed using Snowflake and Power BI to analyze job postings related to data analyst roles. The dataset contains more than two thousand job listings, including information on job titles, company names, industries, locations, salaries, and required technical skills such as SQL, Python, Excel, Tableau, and Power BI.

The project involves extracting and analyzing job market data using SQL queries in Snowflake, identifying patterns in hiring demand, salary distribution, and skill requirements. The processed dataset is then connected to Power BI, where interactive dashboards are created using DAX measures and advanced data visualizations.

The dashboard consists of multiple analytical pages including **Job Market Overview, Salary Insights, Skills Demand Analysis, and Company Hiring Trends**. These dashboards enable users to explore which industries hire the most analysts, which skills are most demanded, which cities offer higher salaries, and which companies actively recruit data analysts.

The results of this analysis reveal that **SQL remains the most demanded technical skill**, Excel continues to be widely required, and Python is associated with higher average salaries. Additionally, industries such as **IT Services and Staffing firms dominate hiring volumes**, while certain cities demonstrate higher salary clusters.

This project demonstrates how business intelligence tools can transform raw job market data into actionable insights that support **career planning, talent development, and hiring strategies**.

DECLARATION

I hereby declare that the project report titled **"Data Analyst Job Market Intelligence Dashboard"** is an original work carried out by me as part of my academic learning and portfolio development.

The work presented in this report has been completed using publicly available job market datasets and analytical tools including Snowflake and Power BI. All analyses, dashboards, and insights presented in this report are the result of my own efforts in data preparation, data modeling, visualization, and interpretation.

This project has not been submitted previously for any academic evaluation or certification.

I take full responsibility for the authenticity and accuracy of the work presented in this report.

Name: Meghana M
Date: 12/03/2026

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1. Project Overview

In the modern digital economy, organizations increasingly rely on data to drive decision-making. As a result, the demand for data analytics professionals has grown significantly across industries. Companies are actively seeking data analysts who possess strong analytical skills along with technical expertise in tools such as SQL, Python, Excel, Tableau, and Power BI.

However, understanding the dynamics of the data analyst job market can be challenging for students, job seekers, and training institutions. Questions such as which skills are most demanded, which industries hire the most analysts, and which cities offer better salaries require systematic analysis of job market data.

The Data Analyst Job Market Intelligence Dashboard project aims to address this challenge by analyzing job posting data and transforming it into meaningful insights using business intelligence tools.

The primary objective of this project is to:

- **Analyze hiring patterns for Data Analyst roles**
- **Identify the most demanded technical skills**
- **Understand salary trends across locations and industries**
- **Discover which companies are actively hiring analysts**
- **Provide actionable insights for job seekers and career planners**

To achieve these objectives, job posting data was analyzed using for SQL-based data exploration and for dashboard creation. Data modeling, transformation, and visualization techniques were applied to convert raw data into an interactive analytical dashboard.

The final output is a multi-page interactive dashboard that enables users to explore hiring trends, skill demand, salary insights, and company hiring behavior in the data analyst job market.

2. Dataset Summary

The dataset used in this project contains approximately 2,000 job postings related to Data Analyst roles collected from online job listings.

The dataset includes several attributes describing different aspects of job postings such as job titles, company names, industry sectors, locations, salary ranges, and required technical skills.

Key Fields in the Dataset

Column	Description
job_title	Title of the job posting
company_name_clean	Name of the hiring company
location_city	City where the job is located
industry	Industry sector of the company
salary	Estimated salary for the role
skill_sql	Indicates whether SQL is required
skill_python	Indicates whether Python is required
skill_excel	Indicates whether Excel is required
skill_tableau	Indicates whether Tableau is required

skill_power_bi	**Indicates whether Power BI is required**

The skill columns are represented as Boolean indicators showing whether a particular skill is required for the job posting.

This structure enables analysis of:

- **Skill demand frequency**
- **Skill combinations required by employers**
- **Relationship between skills and salary**
- **Hiring patterns across industries and locations**

The dataset was initially explored using SQL queries in to understand data distributions and identify patterns.

3. Data Preparation & Modeling

Before building the dashboard, the dataset required preparation and structuring to enable efficient analysis.

Data Cleaning

Data preparation involved several steps including:

- **Removing duplicate records**
- **Standardizing company names**
- **Handling missing salary values**
- **Ensuring consistent industry categories**
- **Converting skill columns into Boolean indicators**

These steps ensured that the dataset was accurate and suitable for analysis.

Feature Engineering

Additional analytical fields were created to support dashboard metrics. For example:

- **Total job count**
- **Average salary calculations**
- **Skill demand counts**
- **Number of companies hiring analysts**
- **Number of cities with job opportunities**

Skill demand was calculated using SQL aggregation queries in Snowflake.

Example logic used:

- **Counting the number of jobs requiring SQL**
- **Counting the number of jobs requiring Python**
- **Comparing demand across multiple skills**

Data Modeling

The dataset was structured using a simple analytical model that supports efficient dashboard calculations.

Key components include:

- **Fact table containing job postings**
- **Dimensions such as company, industry, and location**
- **Skill indicator columns used for skill demand analysis**

4. Job Market Intelligence Analysis

The dashboard is designed to provide a comprehensive view of the data analyst job market through multiple analytical perspectives.

4.1 Job Market Overview

This page provides a high-level summary of the data analyst job market.

Key Metrics Displayed

- **Total Number of Job Postings**
- **Average Salary**
- **Total Hiring Companies**
- **Total Cities with Job Opportunities**
- **Average Jobs per Company**

Visualizations Used

1. **Clustered Bar Chart — Top Cities Hiring Analysts**
2. **Donut Chart — Job Distribution by Industry**
3. **Treemap — Top Companies Hiring Data Analysts**

These visualizations help identify where the majority of job opportunities are located and which industries dominate hiring.

4.2 Salary Insights Analysis

This page focuses on understanding salary patterns across different industries and locations.

Key Analyses

- **Salary distribution across industries**
- **Cities offering higher average salaries**
- **Relationship between technical skills and salary levels**

The dashboard helps identify which industries provide the most competitive compensation for data analysts.

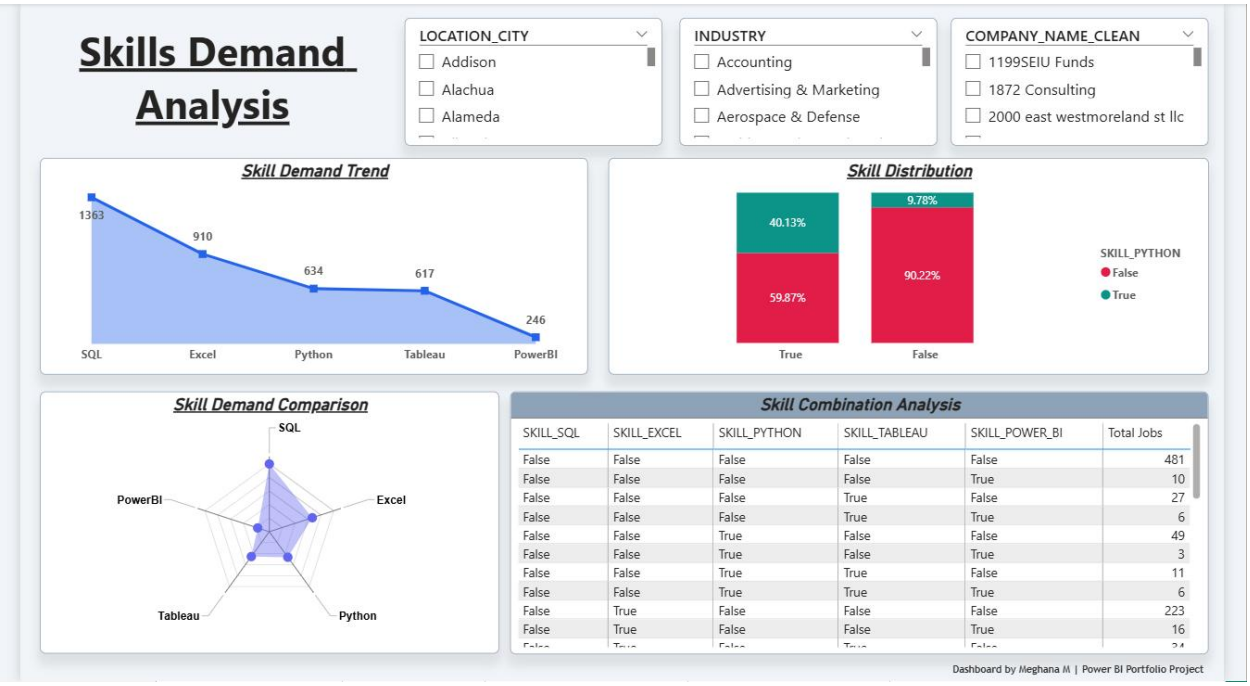
4.3 Skills Demand Analysis

Technical skill demand is a critical factor for aspiring data analysts. This page analyzes the most frequently required skills across job postings.

Key Visualizations

1. **Radar Chart — Skill Demand Comparison**
2. **100% Stacked Column Chart — Skill Distribution**
3. **Area Chart — Skill Demand Trend**
4. **Matrix Visualization — Skill Combination Analysis**

The matrix visualization highlights which combinations of skills commonly appear together in job postings, helping candidates understand which skill sets employers value most.



4.4 Company Hiring Trends

This page analyzes company hiring patterns across industries.

Visualizations

Ribbon Chart — Top Hiring Companies by Industry

This visualization shows how different companies rank within their industries based on the number of job postings.

This analysis helps identify organizations that actively recruit data analysts.

5. Power BI Dashboard Design

The dashboard was built using with a focus on usability, interactivity, and clarity.

Design Principles

The dashboard design followed several principles:

- **Clear page layout and structure**
- **Consistent color themes across pages**
- **Intuitive visualizations for business users**
- **Interactive filters for deeper analysis**

Interactive Features

The dashboard includes multiple slicers such as:

- **City filter**
- **Industry filter**
- **Company filter**

These slicers allow users to dynamically explore job market patterns.

Key Technologies Used

- **Data transformation**
- **DAX measures for KPI calculations**
- **Interactive visualizations**
- **Custom visual elements**

The final result is an interactive dashboard that enables users to explore job market insights efficiently.

6. Key Market Insights

The analysis of the dataset revealed several important insights about the data analyst job market.

Skill Demand Insights

- **SQL is the most commonly required skill in data analyst job postings.**
- **Excel continues to be widely used across many industries.**
- **Python demand is strong, particularly for analytical and automation tasks.**

Industry Insights

- **Technology and IT services companies contribute a large portion of job postings.**
- **Staffing and consulting firms frequently hire analysts to support client projects.**

Geographic Insights

- **Major metropolitan cities show higher concentrations of data analyst job opportunities.**
- **Certain cities also demonstrate higher average salary ranges.**

Hiring Insights

- **A relatively small number of companies account for a significant portion of job postings.**
- **This indicates concentrated hiring activity among large organizations and consulting firms.**

7. Actionable Recommendations

Based on the insights obtained from the analysis, the following recommendations can be made for aspiring data analysts:

Skill Development

Candidates should prioritize learning the following skills:

- **SQL**
- **Excel**
- **Python**
- **Data visualization tools such as Power BI or Tableau**

Skill Combinations

Employers often prefer candidates who possess multiple complementary skills, such as:

- **SQL + Excel**
- **SQL + Python**
- **SQL + Visualization tools**

Location Strategy

Job seekers may benefit from targeting cities with higher hiring activity to increase their job opportunities.

Portfolio Development

Candidates should create practical projects and dashboards to demonstrate their analytical skills to employers.

8. Limitations & Assumptions

Although the project provides valuable insights, certain limitations should be acknowledged.

Data Limitations

- **The dataset represents only a sample of job postings and may not cover the entire job market.**
- **Salary data may contain estimated ranges rather than exact figures.**

Analytical Assumptions

- **Skill indicators assume equal importance across all job postings.**
- **Some job descriptions may include additional skills that are not captured in the dataset.**

Future analysis could include larger datasets and additional skill attributes.

9. Conclusion

The Data Analyst Job Market Intelligence Dashboard successfully demonstrates how data analytics techniques can be used to analyze labor market trends and generate actionable insights.

Using SQL queries in and visualization capabilities of , the project transforms raw job market data into meaningful insights about hiring patterns, skill demand, and salary trends.

The dashboard enables users to explore job market dynamics through interactive visualizations and provides valuable guidance for students and professionals preparing for data analyst roles.

Overall, this project highlights the importance of data-driven decision making in understanding workforce demand and demonstrates practical applications of business intelligence tools in real-world scenarios.